

Alan (Hongyao) Shao

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EDUCATION

Boston University

Bachelor of Science in Data Science & Business Administration

Boston, MA

Expected May 2027

- **GPA:** 3.84 / 4.0; **Honors:** Dean's List, all semesters

DOMAIN KNOWLEDGE & TECHNICAL SKILLS

Domain Knowledge: Data Processing & Visualization, Experimental Design, Statistical Modeling, Web Development, Supervised Machine Learning, Web Scraping, Sentiment Analysis, Feature Engineering

Technical Expertise: Python (Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, SciPy), Rust, Java, LaTeX, Flask, SQL, HTML, JMP, TensorFlow, Tableau

WORK EXPERIENCE

BU Spark!

Technical Project Manager

Boston, MA

Jan 2026 – Present

- Oversaw technical delivery across multiple Spark! student projects by reviewing progress, identifying blockers, and guiding debugging decisions to keep work aligned with client requirements.
- Managed GitHub Pages and organized project artifacts, documentation to preserve technical continuity.

Hongfu Investment

Investment Analyst

Shanghai, China

May 2025 – Aug 2025

- Researched 20+ private technology firms by scraping financial data with Python BeautifulSoup, providing clearer evidence for recommendations.
- Analyzed peer-company financial performance in SQL and Python, comparing revenue CAGR, margins, and valuation multiples to support due diligence on investment targets.

DeFiner Labs

Data Analyst

San Mateo, CA

Dec 2024 – Feb 2025

- Cleaned cryptocurrency and token datasets for market analyses, creating reliable inputs for market monitoring, token evaluation, and user activity tracking.
- Developed Dune Analytics dashboards with SQL to track blockchain activity, turning account-type and transaction-volume trends into weekly market-monitoring updates.

ACADEMIC PROJECTS

Boston Building Heating Systems: BCAN Data Integration

Jan 2026 – May 2026

- Created a building-level dataset for heating-system inference by integrating 3 City of Boston public datasets with Python/pandas, parcel ID matching, and address standardization.
- Screened unstructured permit comments with keyword rules and API-based LLM prompts, identifying heating installation, replacement, upgrade, and fuel-conversion records.
- Designed a client-facing Boston property map with search, filters, and exportable results, helping BCAN identify likely high-emission heating systems for decarbonization outreach.

Progressive Mass: Home Rule Petition

Sep 2025 – Dec 2025

- Implemented an end-to-end pipeline with the MA Legislature Public API to clean and standardize 10 years of Home Rule Petition data into petition details, timelines, and outcomes.
- Mapped petition lifecycles and legislative workflows to identify approval and failure patterns, enabling Progressive Mass to examine petition bias for State House reporting.

Emotion Recognition from Audio Using CNN-Based Spectrograms

Feb 2025 – May 2025

- Developed an audio-processing pipeline for RAVDESS with Python, TensorFlow, and Librosa, converting speech and song files into reusable CNN training inputs.
- Fine-tuned speech and song CNN models by adjusting learning rate, model depth, and batch size, improving F1 score by 50% over baseline and reaching about 80% accuracy.

Delivery Performance Dynamics Analysis

Oct 2023 – Dec 2023

- Tested delivery-performance relationships with OLS regression and ANOVA, turning raw personnel, timing, and satisfaction data into report findings.
- Trained Random Forest models in Python to predict delivery time and identify efficiency drivers, achieving an F1 score above 0.85.